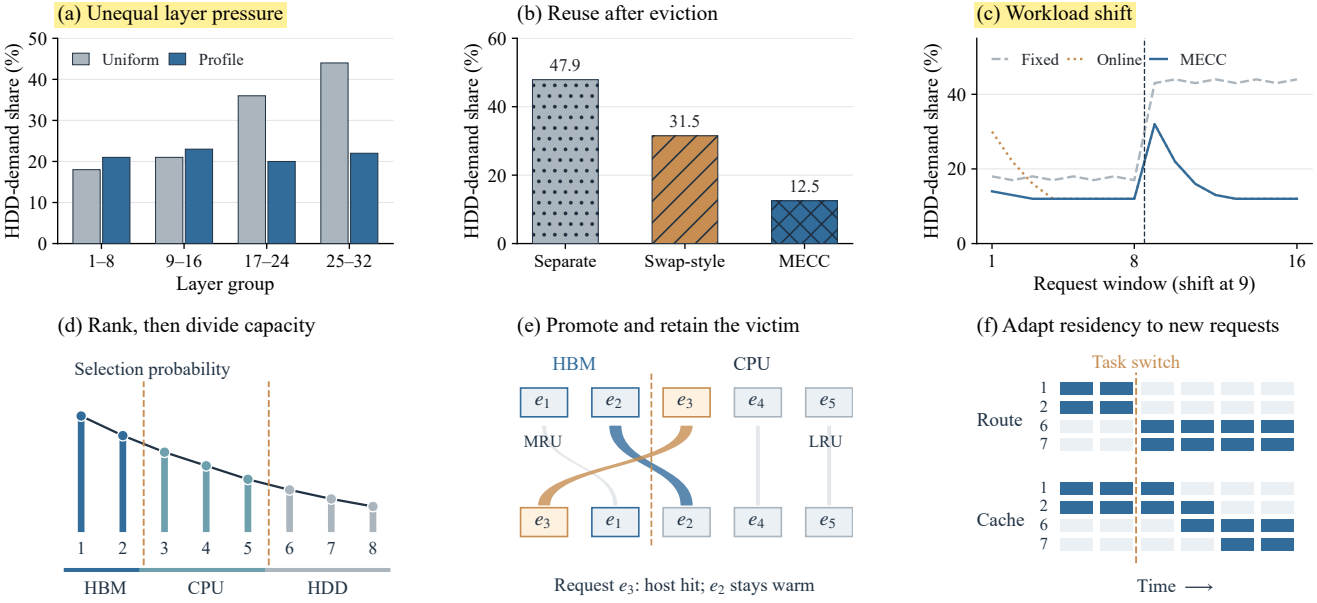

(a) Unequal layer pressure
HDD-demand share (%)
Uniform
Profile
1–8
9–16
17–24
25–32
Layer group
(b) Reuse after eviction
47.9
31.5
12.5
Separate
Swap-style
MECC
(c) Workload shift
Fixed
Online
MECC
Request window (shift at 9)
(d) Rank, then divide capacity
Selection probability
HBM
CPU
HDD
(e) Promote and retain the victim
MRU
LRU
Request e_3: host hit; e_2 stays warm
(f) Adapt residency to new requests
Task switch
Route
Cache
Time